Name: _____ **Roll number:** _____

Gradient boosting, numbers (regression)

Where you see a), b), c): circle the one right answer.

The idea. We want to predict a number (this is called *regression*). We start with one guess for every point. Then we repeat: (1) see what is *left over* (our mistakes); (2) draw a tiny tree that predicts the leftover; (3) add a small part of it to the guess. Every round the guess gets better.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	1	2	3	8	10	12

A **stump** is the smallest tree: one question ("is $x \le t$?") and two leaves. Step size $\nu = \frac{1}{2}$: we add only half of each leaf value.

Formulas you need.
- first guess F_0 = average of y
- leftover $r = y - F$
- leaf value = average of the leftovers in that leaf
- new guess $F_{\text{new}} = F_{\text{old}} + \nu \times$ leaf value
- total leftover = $\sum r^2$ (square each leftover, then add them)

Step 1 First guess and leftover.

(a) The first guess F_0 is the average of y:
a) 5.5 **b)** 6 **c)** 7

(b) The leftover $r = y - F_0$ for points 1 to 6 is:
a) $-5, -4, -3, 2, 4, 6$ **b)** $5, 4, 3, -2, -4, -6$ **c)** $1, 2, 3, 8, 10, 12$

(c) A positive r means our guess is **a)** too high **b)** too low.

Step 2 Choose the split.

A good split puts points with *similar* leftovers in one leaf. Points that satisfy the split (for example $x \le 2.5$) go to the left leaf, the others to the right. Use your leftovers from Step 1(b) and write the ones that fall in each leaf.

split	left leaf	right leaf
$x \le 2.5$		
$x \le 3.5$		
$x \le 4.5$		

Best split:
a) $x \le 2.5$ **b)** $x \le 3.5$ **c)** $x \le 4.5$

Why is it best?

Step 3 Leaf values.

Leaf value = average leftover in the leaf. For the best split, (left, right) = **a)** $-4, +4$ **b)** $-12, +12$ **c)** $+4, -4$ **d)** $-2, +2$

Step 4 Add a small part.

The new guess $F_1 = F_0 + \nu \times$ leaf value, for points 1 to 6, is:
a) 4, 4, 4, 8, 8, 8 **b)** 2, 2, 2, 10, 10, 10 **c)** $-2, -2, -2, 2, 2, 2$

The new leftover $r = y - F_1$ is:
a) $-3, -2, -1, 0, 2, 4$ **b)** $-5, -4, -3, 2, 4, 6$ **c)** $3, 2, 1, 0, -2, -4$

Total leftover $\sum r^2$ (square each leftover, then add) goes from 106 to **a)** 34 **b)** 74 **c)** 106. **Why not add the whole leaf value ($\nu = 1$)?**

Step 5 Round 2.

The new leftovers are $-3, -2, -1, 0, 2, 4$. Choose the split the same way as in Step 2. The better split is **a)** $x \le 3.5$ **b)** $x \le 4.5$. **Why?**

Think. (a) The second tree predicts the **a)** leftover **b)** original y. **Why?**

(b) AdaBoost passes its mistakes to the next tree as: **a)** weights on the points **b)** leftovers
Gradient boosting passes its mistakes to the next tree as: **a)** weights on the points **b)** leftovers

Name: _____ **Roll number:** _____

Gradient boosting, yes/no (classification) **Where you see a), b), c): circle the one right answer.**

The idea. The answer is now yes (1) or no (0). The model keeps a **score** F for every point (any number; $F = 0$ means fifty-fifty) and turns it into a **probability of yes** $p = \sigma(F)$. The loop is the same as on page 1: (1) find the leftover $r = y - p$; (2) draw a stump that predicts the leftover; (3) add a small part of it to the *score*.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	0	0	1	1	0	1

Same stumps as page 1. Step size $\nu = \frac{1}{2}$.

Formulas you need.
- probability of yes $p = \sigma(F) = \dfrac{1}{1+e^{-F}}$; so $F = 0$ gives $p = 0.5$
- first score $F_0 = \ln \dfrac{\text{number of 1s}}{\text{number of 0s}}$
- leftover $r = y - p$
- leaf value $= \sum r \,/\, \sum p(1-p)$, both sums over the points in that leaf
- new score $F_{\text{new}} = F_{\text{old}} + \nu \times$ leaf value
- $\sigma(-1) = 0.269$, $\sigma(0.5) = 0.622$

Step 1 **First score and leftover.**

There are three 1s and three 0s, so $F_0 = \ln \frac{3}{3} =$ **a)** -1 **b)** 0 **c)** 1. Then $p_0 = \sigma(0) =$ **a)** 0 **b)** 0.5 **c)** 1.

The leftover $r = y - p_0$ for points 1 to 6 is:

a) $-0.5, -0.5, 0.5, 0.5, -0.5, 0.5$ **b)** $0.5, 0.5, -0.5, -0.5, 0.5, -0.5$ **c)** 0, 0, 1, 1, 0, 1

Step 2 **Choose the split.**

A good split puts the 1s on one side and the 0s on the other, as far as possible. For each split write the y values of the points that go left and right.

split	left leaf (y values)	right leaf (y values)
$x \le 2.5$		
$x \le 3.5$		
$x \le 4.5$		

Best split:
a) $x \le 2.5$ **b)** $x \le 3.5$ **c)** $x \le 4.5$

Why is it best?

Step 3 **Leaf values.**

Every point has $p_0(1-p_0) = 0.25$. For the best split fill in:

leaf	$\sum r$	$\sum p(1-p)$	leaf value $= \sum r / \sum p(1-p)$
left leaf			
right leaf			

Step 4 **Add a small part, to the score.**

Use $F_1 = F_0 + \nu \times$ leaf value, then $p_1 = \sigma(F_1)$ (use the σ values in the formula box), then $r_1 = y - p_1$ (round to 3 decimals).

point i	1	2	3	4	5	6
F_1						
p_1						
$r_1 = y - p_1$						

Step 5 **What did the first tree miss?**

Which point has the biggest leftover now (ignoring the sign)? **a)** point 1 **b)** point 3 **c)** point 5. **Why?**

Think. Why do we add the leaf value to the score F, and not directly to the probability p?

(b) A point has $p = 0.5$ and $y = 1$. Is its leftover r **a)** positive **b)** negative, so does the next tree push its score **a)** up **b)** down?

Name: _____ **Roll number:** _____

AdaBoost, yes/no (classification)

Where you see a), b), c): circle the one right answer.

The idea. The answer is a class (*classification*). A *judge* is a stump: one question about x; it answers $+1$ or -1. We build a committee, one judge per round. After each round the points the judge got **wrong** count **more**, so the next judge pays attention to them. A judge with few mistakes gets a **louder voice** α. The committee's answer is the **sign** of the total vote.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	$+1$	$+1$	-1	-1	$+1$	-1

Three judges (each says $+1$ if $x \le t$, else -1)
A: $t = 2.5$ B: $t = 4.5$ C: $t = 5.5$

Formulas you need.
- mistake score ε = sum of the weights of the points the judge gets wrong
- voice $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$
- new weight = old $\times e^{\alpha}$ (point was wrong) or old $\times e^{-\alpha}$ (point was right), then divide by the total so the weights add to 1
- vote $F(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x)$, where $h_k(x) = \pm 1$ is what judge k answers; the committee's answer $= +1$ if $F > 0$, -1 if $F < 0$
- $\ln 5 = 1.609$, $\ln 4 = 1.386$, $e^{0.805} = 2.236$, $e^{-0.805} = 0.447$

Step 1 Pick the judge.

At the start every point has weight **a)** 1/2 **b)** 1/6 **c)** 1. So ε = (number of points wrong)/6. The best judge is the one with the smallest ε.

judge	rule	points it gets wrong	ε
A	$x \le 2.5 \to +1$, else -1		
B	$x \le 4.5 \to +1$, else -1		
C	$x \le 5.5 \to +1$, else -1		

Best judge: **a)** A **b)** B **c)** C
$\varepsilon_1 =$ **a)** 1/6 **b)** 2/6 **c)** 3/6

Step 2 Give the judge a voice.

$\alpha_1 = \frac{1}{2} \ln \frac{1-\varepsilon_1}{\varepsilon_1} =$ **a)** 0.805 **b)** 1.609 **c)** 0.5

If $\varepsilon = \frac{1}{2}$ (a coin toss), then $\alpha =$ a) 0 **b)** 1 **c)** 0.5. **What does that tell you about such a judge's vote?**

Step 3 Change the weights.

Judge A got point **a)** 3 **b)** 4 **c)** 5 wrong. Work out the new weights yourself (3 decimals; tiny rounding differences are fine), using $e^{\alpha} = 2.236$ and $e^{-\alpha} = 0.447$ from the formula box.

	old weight $\times$ factor =	new weight (before dividing)
the wrong point	$\frac{1}{6} \times 2.236 =$	
each right point (5 of them)	$\frac{1}{6} \times 0.447 =$	
total of all six	(wrong point) + 5$\times$ (right point) =	
after dividing by the total	wrong point:	right point:

Why do we deliberately make the wrong point heavier?

Step 4 Round 2.

Now point 5 has weight 0.5 and every other point 0.1. Find ε for each judge; add the weights of the points it gets wrong (same wrong points as Step 1).

judge	rule	wrong on points	ε
A	$x \le 2.5$		
B	$x \le 4.5$		
C	$x \le 5.5$		

Best judge: **a)** A **b)** B **c)** C
$\alpha_2 =$ **a)** 0.693 **b)** 0.805 **c)** 1.386

Judge A was best in round 1. Why is it not chosen again?

Step 5 Take the vote.

$F(x) = 0.805\, h_1(x) + 0.693\, h_2(x)$ (h_1: judge A, h_2: judge C). For point 5:

$h_1(5) =$ **a)** $+1$ **b)** -1 $h_2(5) =$ **a)** $+1$ **b)** -1 $F =$ **a)** -0.112 **b)** $+0.112$ **c)** $+1.498$

The committee answers **a)** $+1$ **b)** -1. The true $y_5 = +1$, so the committee is **a)** right **b)** wrong.

Think. Why must every judge be better than a coin toss ($\varepsilon < \frac{1}{2}$)?

Name: _____ **Roll number:** _____

AdaBoost, numbers (regression)

Where you see a), b), c): circle the one right answer.

The idea. Same committee idea as page 3, but now a point is not just right or wrong: a tree can be wrong by a *little* or by a *lot*. So we measure **how** wrong each point is, compared with the worst mistake of the round. Points predicted badly get **relatively more weight** next round. A tree with a small average loss gets a **louder voice** α. The final answer is the **weighted median** of the trees' answers.

point i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	2	2	2	6	6	10

Round 1: every weight is $\frac{1}{6}$. The best question is $x \le 3.5$: points 1 to 3 go to the left leaf, points 4 to 6 to the right leaf.

Formulas you need.
- leaf value h = average of y in the leaf (all weights are equal in round 1)
- error $\delta = |y - h|$; Δ = the largest δ of the round; relative loss $\ell = \delta/\Delta$ (between 0 and 1)
- average loss $\bar{\ell} = \sum w\,\ell$ (the weights add to 1); $\beta = \bar{\ell}\big/(1-\bar{\ell})$; voice $\alpha = \ln(1/\beta)$
- new weight $\propto w \times \beta^{\,1-\ell}$, then divide by the total so the weights add to 1
- weighted median: sort the answers, add up their α from the smallest answer, stop at the first answer where the running total reaches half of all the α
- $\ln 2 = 0.693$, $\sqrt{0.5} = 0.707$

Step 1 Grow the tree and measure the loss.

Work to 2 decimals; tiny rounding differences are fine.

Left leaf $h =$ Right leaf $h =$ Largest error $\Delta =$

point i	1	2	3	4	5	6
y	2	2	2	6	6	10
h						
$\delta = \lvert y - h \rvert$						
$\ell = \delta/\Delta$						

Step 2 Give the tree a voice.

$\bar{\ell} = \frac{1}{6}(\ell_1 + \cdots + \ell_6) =$ $\beta = \bar{\ell}/(1-\bar{\ell}) =$ $\alpha = \ln(1/\beta) =$

A tree whose average loss $\bar{\ell}$ is close to $\frac{1}{2}$ (almost a coin toss) gets a **a)** loud **b)** quiet voice. **Why?**

Step 3 Change the weights.

Multiplier $= \beta^{1-\ell}$ (a perfect point, $\ell = 0$, gets β^1; the worst point, $\ell = 1$, gets $\beta^0 = 1$). New weight = multiplier ÷ (sum of the six multipliers).

point i	1	2	3	4	5	6
ℓ	0	0	0	0.5	0.5	1
multiplier						
new weight						

sum of multipliers =

The point with the largest weight is **a)** point 1 **b)** point 4 **c)** point 6. **Why do the perfectly predicted points lose weight?**

Step 4 The final answer.

At $x = 6$ three trees answer as below. Find the weighted median.

tree	1	2	3
answer at $x = 6$	7.33	9.0	5.0
voice α	0.69	0.90	0.41

Sorted answers:

Running total of α:

Half of all $\alpha =$

The weighted median is **a)** 5.0 **b)** 7.33 **c)** 9.0. **Why a median and not an average?**

Think. On page 3 a judge needs $\varepsilon < \frac{1}{2}$. What is the matching condition here? **a)** $\bar{\ell} < \frac{1}{2}$ **b)** $\bar{\ell} > \frac{1}{2}$.